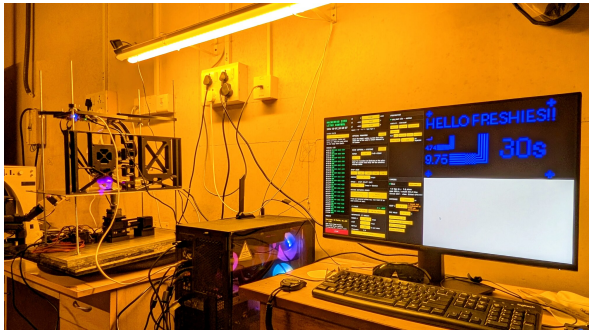
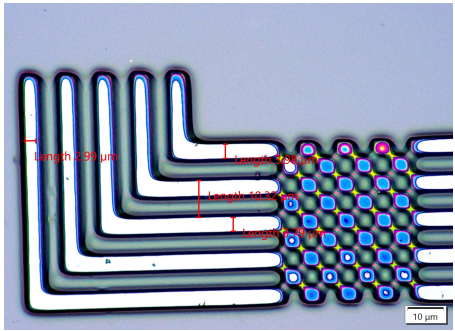
Design and Development of a <$2000 Direct-Write Photolithography System



Direct-write photolithography setup



3 µm lines patterned by our system

HackerFab IITB

democratizing semiconductor fabrication

hackerfabiitb.github.io
hackerfabiitb@gmail.com

1. **Who we are:** A student-led R&D project guided by IIT Bombay SemiX faculty. We are building low-cost open-source student-accessible tools for making chips and transistors.
2. **Our Mission:** To develop a low-cost DIY semiconductor fab capable of making working transistors, whose open-source recipe can be replicated and developed by colleges across India.
3. **Our Vision:** To empower every EE student in India to learn to fabricate semiconductor devices (like a Tinkerer's Lab or Makerspace for semiconductors)
4. **Advised by SemiX faculty:** Prof. Swaroop Ganguly, Prof. Udayan Ganguly, Prof. Anil Kottantharayil and Prof. Veeresh Deshpande
5. **Recognition:**
   - 5.1 won User Design Track at 39th International Conf. On VLSI Design for our DIY toolkit & **Fab-in-a-Box** proposal
   - 5.2 Interview and article by Swarajya Magazine[1]
   - 5.3 Deep-dive video by RuntimeBRT[2]

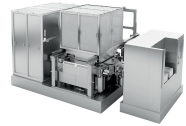


Range of Lithography Systems

Sr No	Tool Name	Resolution (nm)	Throughput (wafers/hr)	Cost (USD)
1	ASML TWINSCAN EXE:5200B High-NA EUV scanner[3, 4]	8	175	$380M
2	IMS MBMW-101 / -301 multi-beam mask writer[5, 6, 7]	14	0.1 *masks*/hr	$80M
3	Raith VOYAGER 50 kV e-beam lithography[8]	8	0.02	$5M
4	Durham Magneto Optics MicroWriter ML3 Pro maskless laser direct-write[9, 10]	600	0.4	$0.5M
5	**HackerFab IITB Litho** DMD projection direct-write	3000	0.003	**$2k**



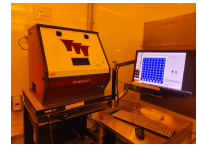
1. ASML EXE:5200B



2. IMS MBMW-101



3. Raith VOYAGER

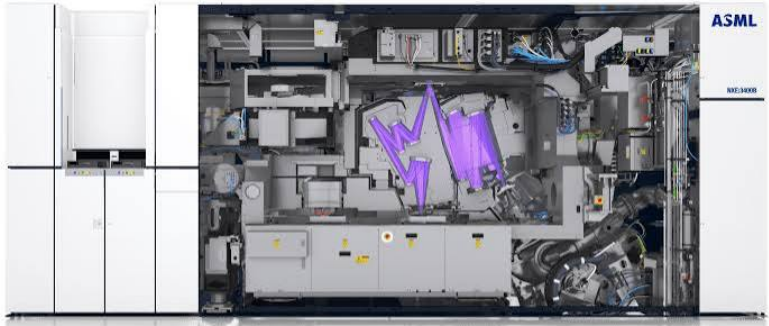


4. MicroWriter ML3



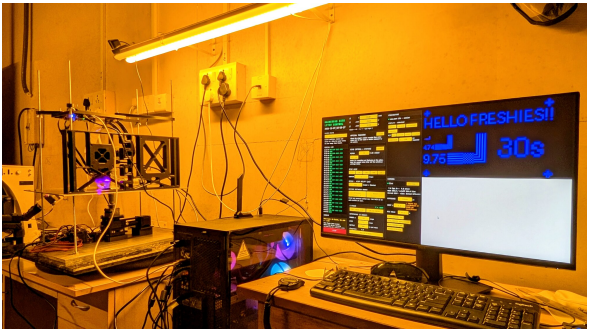
5. HackerFab Litho

ASML EUV

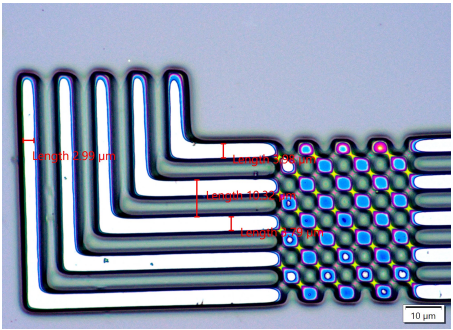


ASML NXE:3400B EUV scanner (CNBC)[11]

HackerFab Litho

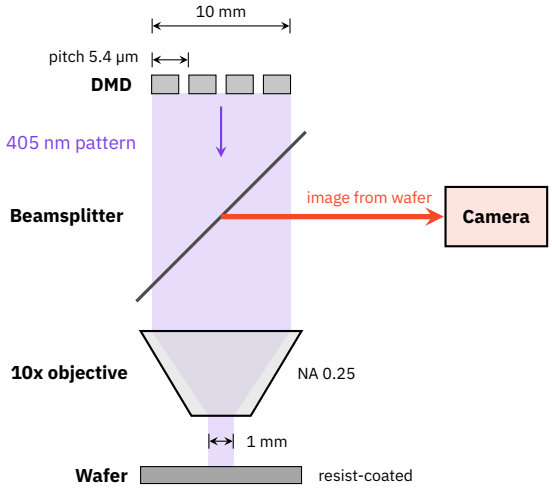


Direct-write photolithography setup



3 µm lines patterned by our system

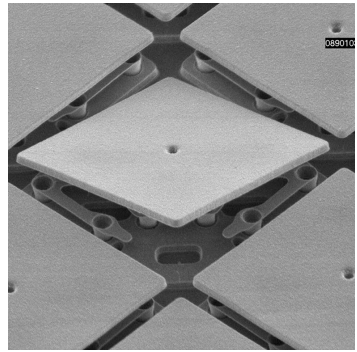
Optical Diagram



Digital Micromirror Device (DMD)



TI DLPDLCR471TPEVM evaluation module



Close-up of DMD micromirrors (IEEE Spectrum)[12]

DLP471TP DMD[13]: **5.4 µm mirror pitch**, 1920 × 1080 mirrors, ±17° tilt, 0.47” diagonal (10.37 × 5.83 mm)

With ≈10x demagnification: ≈0.54 µm pixel and ≈1.0 × 0.58 mm field on the wafer

Optical Specs

1. **Objective:** 10X DIN plan objective[14]
   1.1 NA = **0.25**
   1.2 EFL = 15.37 mm, WD = 1.5 mm
2. **Wavelength:** λ = **405 nm**
3. **Diffraction limit:**
   3.1 Airy disk diameter: $1.22\,\lambda/\mathrm{NA}$ = 1.98 µm
   3.2 Rayleigh resolution: $0.61\,\lambda/\mathrm{NA}$ = **0.99 µm**
4. **Measured:** 3 µm lines, so $k_1 = \mathrm{CD}\cdot\mathrm{NA}/\lambda \approx 1.85$. The optics are not what limits us.

Is $1.22\,\lambda/\mathrm{NA}$ the right formula?

1. $1.22\,\lambda/\mathrm{NA}$ is the *diameter* of the Airy disk, out to the first dark ring.
2. Two points are just resolved when one sits on the other's first dark ring (Rayleigh criterion), so the resolution is the *radius*: $0.61\,\lambda/\mathrm{NA}$.
3. Lithography writes this as $\mathrm{CD} = k_1\,\lambda/\mathrm{NA}$.

What sets the 1.22?

1. **Pupil shape:** $1.22 = 3.83/\pi$, the first zero of the Bessel function J_1, for a uniformly lit *circular* aperture. A slit gives 1.0.
2. **Criterion:** Rayleigh 0.61, Abbe 0.5, Sparrow 0.47.
3. **Illumination coherence (σ):** coherent light resolves line/space down to a half-pitch of $0.5\,\lambda/\mathrm{NA}$; off-axis illumination gets down to $0.25\,\lambda/\mathrm{NA}$.
4. **In practice:** aberrations, defocus, the resist and the process push k_1 higher.

How we built Litho

Q] Why only a 10x objective?

Why not use a 100x lens and get 500 nm resolution?

Characterising Performance of a Litho System

1. Critical Dimension (CD) / Minimum Feature Size / Resolution
2. Line/Space
3. Latent Image
4. Developed Image

Thank you!

Questions?



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[8] Raith, “VOYAGER: e-beam lithography system.” https://raith.com/products/voyager/

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[14] Edmund Optics, “10X DIN Plan Commercial Grade Objective.”
https://www.edmundoptics.in/p/10x-din-plan-commercial-grade-objective/5386